

PROBLEM STATEMENT

The Water Jug Problem is a classical Artificial Intelligence problem that involves two water jugs with fixed capacities and an unlimited water source. The objective is to measure a specific quantity of water using only the available jugs and the allowed operations. The permitted operations include filling a jug completely, emptying a jug, and pouring water from one jug to another until either the source jug becomes empty or the destination jug becomes full. The problem requires finding a sequence of valid operations that results in the desired amount of water.